Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Eksamen VRAESTEL

Kopiereg voorbehou

Exam QUESTION PAPER

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2018EBN111E1

Elektrisiteit en Elektronika

June 2018

2018EBN111E1

Electricity and Electronics

June 2018

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

MEMO

Eksamensinligting: *Exam information:*			
Maksimum punte: *Maximum marks:*	92	Volpunte: *Full marks:*	90
Duur van vraestel: *Duration of paper:*	190 min *190 min*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total numer of pages (including this page):*		35 + 2 OCR answer pages	

BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die Antwoordvorm ingevul word.
   All questions must be answered on the Answer form.
3. Beide die vraestel en die antwoordblad word aan die einde van die eksamen ingeneem.
   Both the question paper and the answer sheet will be taken in at the end of the exam.

Dosent(e): *Lecturer(s):*	1. D le Roux
	2. D Ramotsoela
	3. X Ye
Interne moderator(e): *Internal moderator(s):*	J. Joubert

INSTRUCTIONS

Do step 0 before the start of the session.
Do step 1 in the first 180 minutes of the test
Do step 2 within the following 20 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 180 min):

- Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test.
- Write down the answers for each question on the PRACTICE ANSWER SHEET at the end of the QUESTION PAPER. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (180 min to 200 min):

Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN111E1**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- Do not use commas! Please make use of full stops.

- Complex numbers:

Rectangular form: always place the "j" **at the start** of the imaginary part of the complex number.
Polar form: always add the " ° " (degree sign) after the angle, and make a clear angle sign.

Examples

01	–	1	5	4	.	9				k	Ω							
02	1	.	2							r	a	d	÷	s				
03	–	1	.	4	+	j	8			μ	A							
04	1	4	0	°														
05	3	4	0	∠	4	5	°			V								
06	–	j	5	4						k	Ω							
07	1	∠	2	0	°													
08	5	0	∠	–	1	3	0	°		m	A							

SECTION / AFDELING 1 [46]

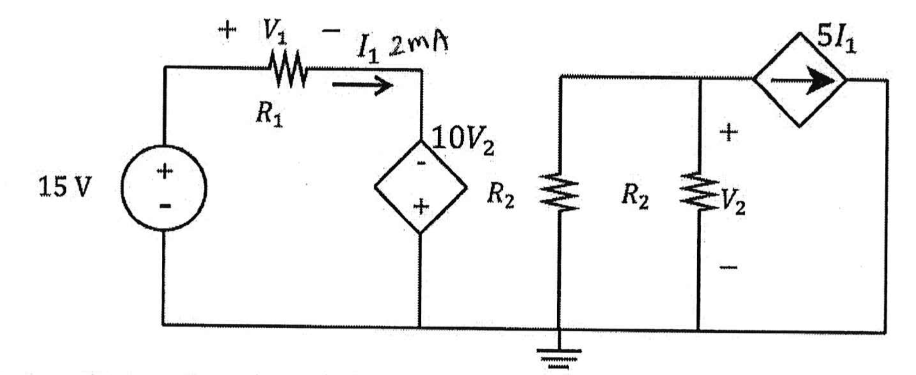
Question 1 & 2 [4]			
01	A power supply with a terminal voltage of t^2 *V* delivers a constant current of *2 A* to a factory machine. Determine how long the machine was on for if it is known that it consumed *144 kJ* of energy during this period.	**01**	'n Kragbron met 'n terminaalspanning van t^2 *V* voorsien 'n konstante stroom van *2 A* aan 'n fabrieksmasjien. Bepaal hoe lank die masjien aangeskakel was indien dit in die periode *144 kJ* se energie verbruik het.
02	It takes 40 minutes to bake 4 trays of muffins in a 7 kW oven. If each tray has a capacity of 9 muffins, how much does it cost to bake 180 muffins? (Assume electricity costs 15 cents/kWh and that the oven is always used at maximum capacity.)	**02**	Dit neem 40 minute om 4 skinkborde se kolwyntjies in 'n 7 kW oond te bak. Indien elke skinkbord 9 kolwyntjies kan vat, hoeveel kos dit om 180 kolwyntjies te maak? (Aanvaar elektrisiteit kos 15 sent/kWh en dat die oond altyd op maksimum kapasiteit funksioneer?

01) $W = \int_0^t 2t^2\,dt = \frac{2}{3}t^3\Big|_0^t = 144\times10^3$

$\Rightarrow t = 60s$

02) $Cost = 0.15 \times \frac{180}{9\times4} \times 7 \times \frac{40}{60} = R\,3.50$

Question 3 & 4 [4]			
03	Find the power associated with the dependent voltage source if $I_1 = 2\ mA$ and $R_2 = 2\ k\Omega$.	**03**	Bepaal die drywing geassosieer met die afhanklike spanningsbron, indien $I_1 = 2\ mA$ en $R_2 = 2\ k\Omega$.
04	Find the power associated with resistor R_1 if $V_2 = 0.5\ V$ and $R_1 = 1\ k\Omega$.	**04**	Bepaal die drywing geassosieer met resistor R_1 indien $V_2 = 0.5\ V$ en $R_1 = 1\ k\Omega$.



03: $I_1 = 2\,mA, \quad 5I_1 = 10\,mA$

$V_2 = -5\,mA \cdot 2\,k\Omega = -10\,V.$

$P = -10V_2 \cdot I_1 = -(10 \cdot (-10)) \cdot 2(mA) = \underline{0.2\,W}$

04: $P_{R_1} = I_1 \cdot V_1 \quad ; \quad 10V_2 = 5V.$

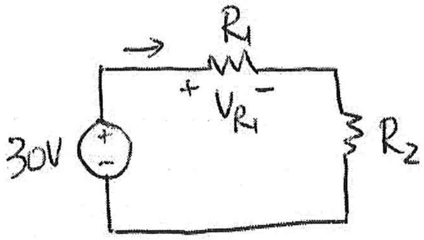
$V_1 = 15 + 5 = 20V$

$I_1 = \dfrac{15+5}{1K} = 20\,mA$

$\Rightarrow P_{R_1} = 20 \times 20 \times 10^{-3} = \underline{0.4\,W}$

Question 5 [2]			
05	A 30V voltage source is connected in series with 2 resistors (R_1 and R_2). Calculated the voltage V_{R1} across resistor R_1 if the resistance of R_1 is half the resistance of R_2.	**05**	'n 30 V spanninsbrong word in series met 2 resistors (R_1 en R_2) gekoppel. Bereken die spanning V_{R1} oor resistor R_1 indien die weerstand van R_1 die helfte van die weerstand van R_2 is.

05:

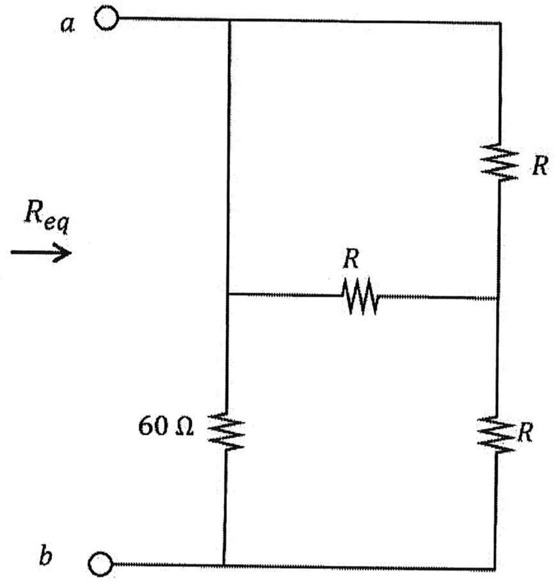


$$R_1 = \frac{1}{2} R_2 \Rightarrow V_{R_1} = \frac{1}{2} V_{R_2}$$

$$\Rightarrow V_{R_2} = 2\, V_{R_1}$$

$$\Rightarrow 3 V_{R_1} = 30 \Rightarrow \underline{V_{R_1} = 10\,V}$$

Question 6 [2]			
06	Find R if $R_{eq} = 30\ \Omega$.	06	Bepaal R indien $R_{eq} = 30\ \Omega$.



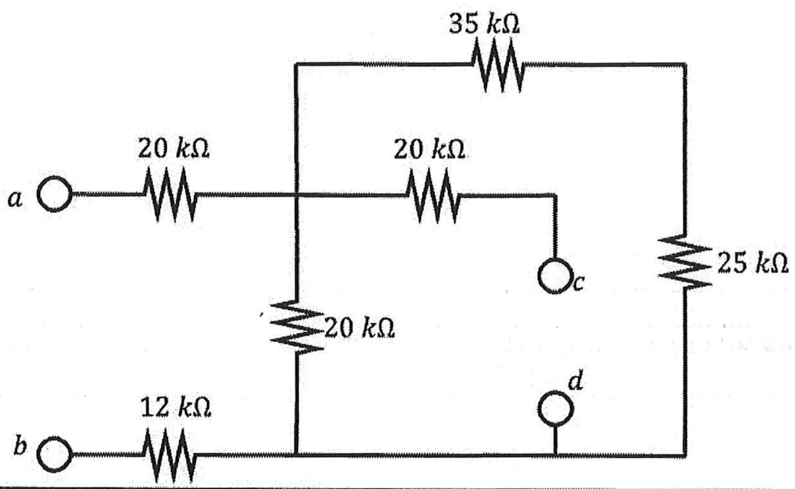
$$R_{eq} = 60 \,||\, (R||R + R)$$

$$= 60 \,||\, 1.5R$$

$$= \frac{60 \times 1.5R}{60 + 1.5R} = 30$$

$$\Rightarrow R = 40\,\Omega.$$

Question 7 [2]			
07	Calculate the equivalent resistance R_{cd} w.r.t. terminals *c-d*.	07	Bepaal die ekwivalente weerstand R_{cd} m.v.t. terminale *c-d*.

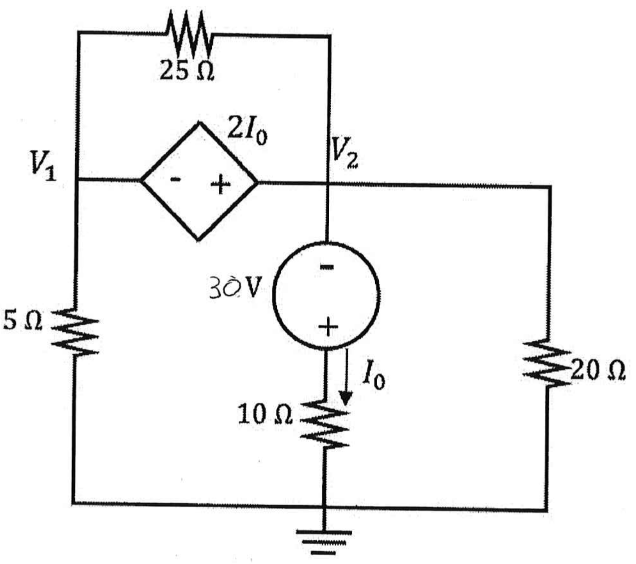


$$R_{cd} = 20 + 20 \,||\, (25+35)$$

$$= 20 + \frac{20 \times 60}{20+60} = 35 k\Omega.$$

Question 8-11 [4]	
Use nodal analysis to set up two equations of the following form: $aV_1 + bV_2 = -60$ (Node 1) $cV_1 + dV_2 = 30$ (Node 2)	Gebruik node analise om twee vergelykings van die volgende vorm op te stel: $aV_1 + bV_2 = -60$ (Node 1) $cV_1 + dV_2 = 30$ (Node 2)

08	*a*	**09**	*b*	**08**	*a*	**09**	*b*
10	*c*	**11**	*d*	**10**	*c*	**11**	*d*



$V_1 + 2I_0 = V_2$, $I_0 = \frac{V_2+30}{10} = 0.1V_2 + 3$

$V_1 + 0.2V_2 + 6 = V_2$

$\Rightarrow V_1 - 0.8V_2 = -6 \Rightarrow 10V_1 - 8V_2 = -60$

$\hookrightarrow -5V_1 + 4V_2 = 30$

$\frac{V_1}{5} + \frac{V_2+30}{10} + \frac{V_2}{20} = 0$

$\Rightarrow 4V_1 + 2V_2 + 60 + V_2 = 0$

$4V_1 + 3V_2 = -60 \Rightarrow -2V_1 - 1.5V_2 = 30$

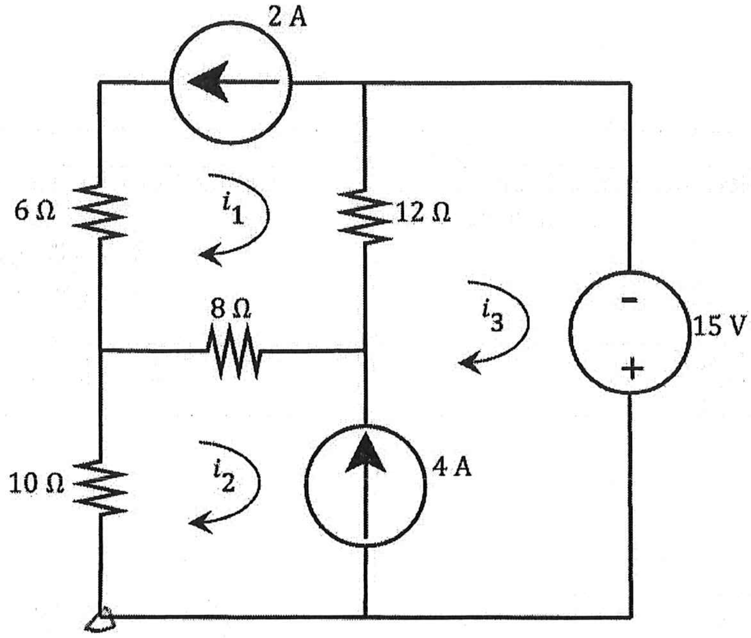
$\hookrightarrow 4V_1 + 3V_2 = -66$

$a = 10$, $b = -8$, $c = -2$, $d = -1.5$

$a = -5$ $b = +4$ $c = 4$ $d = 3$

Question 12-15 [4]	
Use mesh analysis to set up two equations of the following form: $$ei_2 + 12i_3 = f$$ $$gi_2 + hi_3 = 4$$	Gebruik lusanalise om twee vergelykings van die volgende vorm op te stel: $$ei_2 + 12i_3 = f$$ $$gi_2 + hi_3 = 4$$

12	e	13	f	12	e	13	f
14	g	15	h	14	g	15	h



$i_1 = -2$, $i_3 - i_2 = 4 \Rightarrow -i_2 + i_3 = 4$;

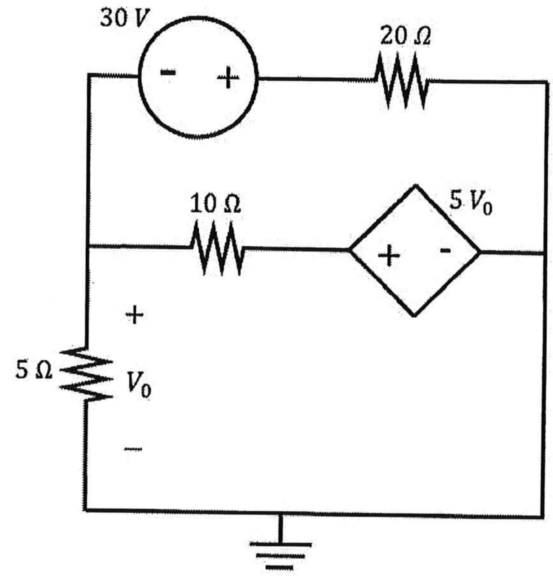
$10(i_2) + 8(i_2 + 2) + 12(i_3 + 2) - 15 = 0$

$\Rightarrow 18i_2 + 12i_3 = -25$

$\Rightarrow e = 18$; $f = -25$;

$g = -1$; $h = 1$

Question 16 [2]			
16	Calculate V_0 using either nodal or mesh analysis.	**16**	Bereken V_0 deur lus- of node-analise te gebruik.



$$\frac{V_0}{5} + \frac{V_0 - 5V_0}{10} + \frac{V_0 + 30}{20} = 0$$

$$\Rightarrow 4V_0 + 2V_0 - 10V_0 + V_0 + 30 = 0$$

$$-3V_0 = -30$$

$$\Rightarrow V_0 = 10\,V.$$

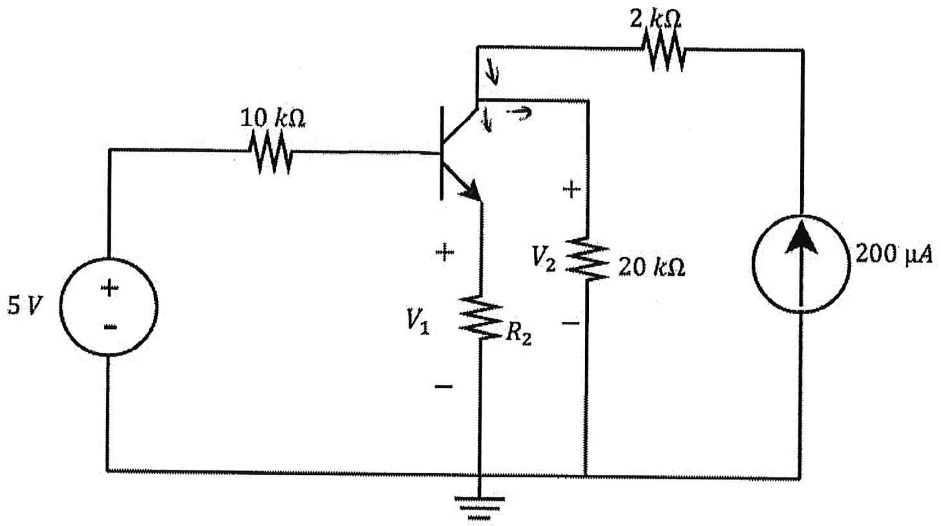
Question 17 [2]			
17	Find the determinant of matrix M below: $$M = \begin{bmatrix} 2 & 0 & 4 \\ 1 & 2 & 3 \\ 4 & 6 & 2 \end{bmatrix}$$	17	Bepaal die determinant van matriks M: $$M = \begin{bmatrix} 2 & 0 & 4 \\ 1 & 2 & 3 \\ 4 & 6 & 2 \end{bmatrix}$$

$$\Delta_M = \left|\begin{array}{ccc|cc} 2 & 0 & 4 & 2 & 0 \\ 1 & 2 & 3 & 1 & 2 \\ 4 & 6 & 2 & 4 & 6 \end{array}\right|$$

$$= 8 + 24 - 32 - 36$$

$$= -36$$

Question 18 & 19 [4]			
Assume that $\beta = 99$ and $V_{BE} = 0.7$ V.		Aanvaar $\beta = 99$ en $V_{BE} = 0.7$ V.	
18	Determine V_1 if $R_2 = 300\ \Omega$.	18	Bepaal V_1 indien $R_2 = 300\ \Omega$.
19	Determine V_{CE} if $R_2 = 0\ \Omega$ and $I_B = 1.25\ \mu A$.	19	Bepaal V_{CE} indien $R_2 = 0\ \Omega$ en $I_B = 1.25\ \mu A$.



18: $-5 + 10\times10^3 i_B + 0.7 + (\beta+1) i_B \cdot R_2 = 0$

$V_1 = (\beta+1) i_B \cdot R_2 = 100 i_B \cdot 300$

$\Rightarrow i_B = 1.\cancel{075\,mA}$ $10.75\,mA$

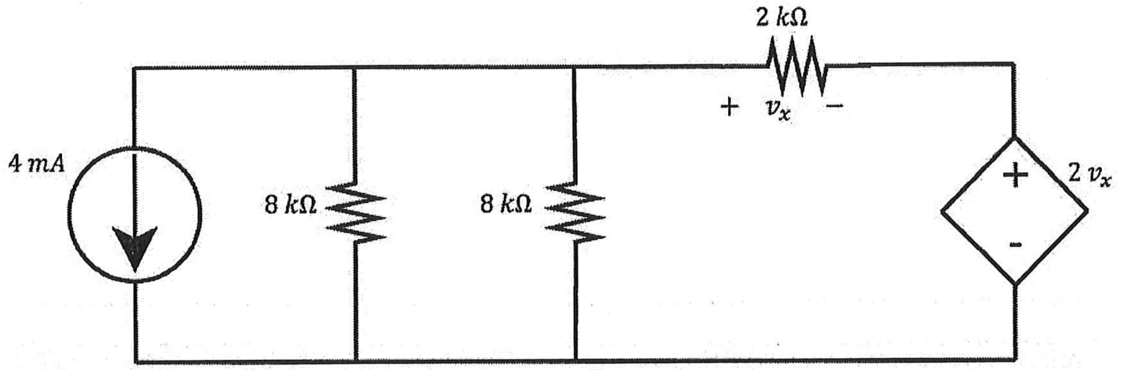
$\Rightarrow \underline{V_1 = 3.225\,V}$

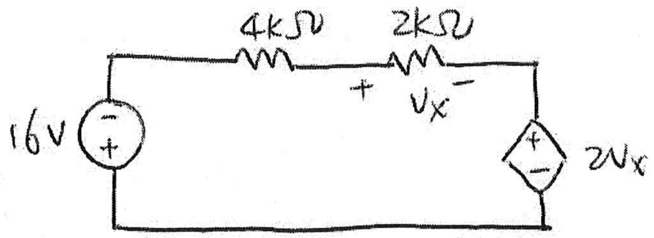
19: $V_{CE} = V_2 = 20k \cdot [200\mu A + (-\beta i_B)]$

$= 20\times10^3 \cdot (200\times10^{-6} - 99\times1.25\times10^{-6})$

$\underline{= 1.525V}$

Question 20 [2]			
20	Use source transformation to determine v_x.	20	Gebruik brontransformasie om v_x te bereken.

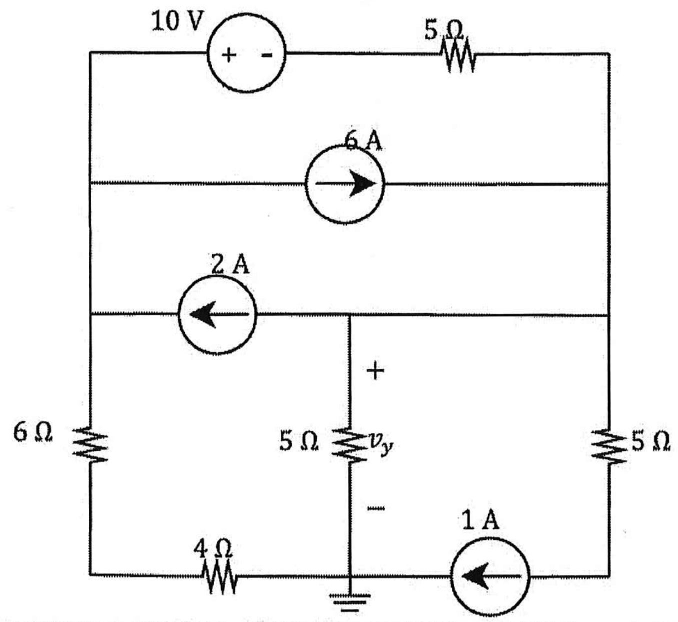




$$16 + 2V_x + V_x + 2V_x = 0$$

$$\Rightarrow V_x = -3.2V.$$

Question 21 & 22 [4]			
21	Find the exclusive contribution of the 1 A current source to v_y.	**21**	Bepaal die ekslusiewe bydrae van die 1 A stroom-bron tot v_y.
22	Find the exclusive contribution of the 10 V voltage source to v_y.	**22**	Bepaal die ekslusiewe bydrae van die 1 A spannings-bron tot v_y.

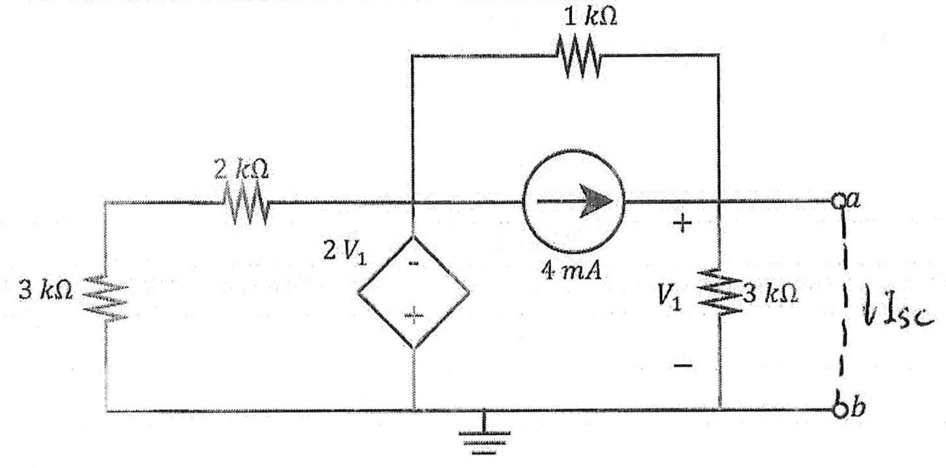


21: when only (1A) left, then

$$V_y^{1A} = -5\cdot 1\cdot \frac{6+4+5}{6+4+5+5} = -5\cdot\frac{15}{20} = -3.75\,V.$$

22: $$V_y^{10V} = -\frac{10}{20}\cdot 5 = -2.5\,V.$$

Question 23 & 24 [4]			
Find the Thevenin equivalent w.r.t. terminals *a-b*.		Bepaal die Thevenin ekwivalent m.v.t. terminale *a-b*.	
23	V_{Th}	23	V_{Th}
24	R_{Th}	24	R_{Th}

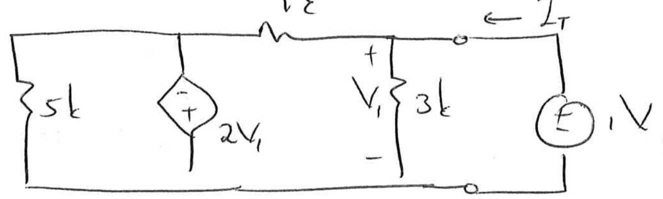


23: $V_{Th} = V_1$, nodal analysis

$$-4\times10^{-3} + \frac{V_1}{3K} + \frac{V_1 + 2V_1}{1K} = 0$$

$$\Rightarrow -12 + V_1 + 9V_1 = 0 \quad \Rightarrow V_1 = V_{Th} = 1.2\,V$$

24: $I_{sc} = 4\,mA$, $R_{Th} = \dfrac{V_{Th}}{I_{sc}} = 300\,\Omega$.



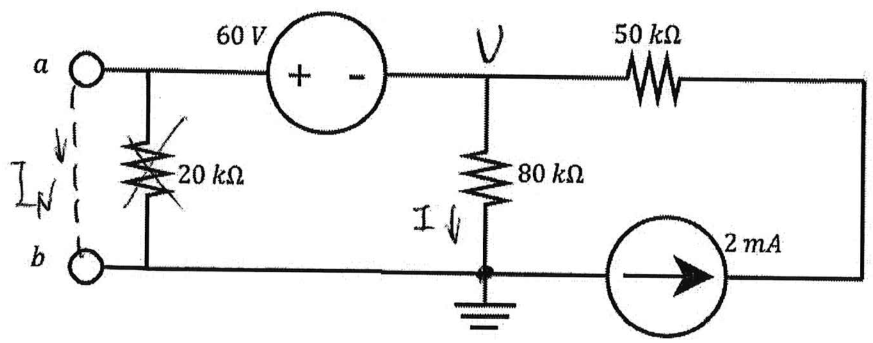
$$I_T = \frac{1}{3k} + \frac{1-(-2)}{1k}$$

$$= \frac{1}{3k} + \frac{3}{1k}$$

$$= \frac{10}{3k}$$

$$\therefore R_{TH} = \frac{1}{I_T} = \frac{3k}{10} = \underline{300\,\Omega}$$

Question 25 & 26 [4]			
Find the Norton equivalent w.r.t. terminals *a-b*.		Bepaal die Norton ekwivalent m.v.t. terminale *a-b*.	
25	I_N	25	I_N
26	R_N	26	R_N

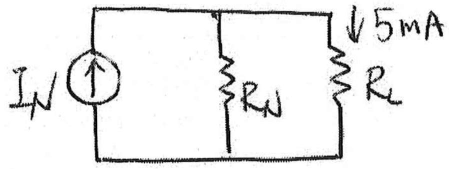


$R_N = 20 \| 80 = 16 k\Omega.$

$V = -60V, \quad I = \frac{-60}{80k} = -0.75 \, mA$

$I_N = 2mA - I = 2.75 \, mA.$

Question 27 & 28 [2]			
Assume that a Norton equivalent circuit is connected to a load resistor R_L and that maximum power p_{max} is delivered to the load. If the current through R_L is 5 mA and p_{max} = 125mW, find the Norton equivalent current I_N and resistance R_N		Aanvaar dat 'n Norton ekwivalente kringbaan aan 'n lasweerstand R_L gekoppel is, en dat maksimum drywing p_{max} aan die las gelewer word. Indien die stroom deur R_L gelyk is aan 5 mA, en p_{max} = 125mW, bepaal die Norton ekwivalente stroom I_N en weerstand R_N	
27	I_N	27	I_N
28	R_N	28	R_N



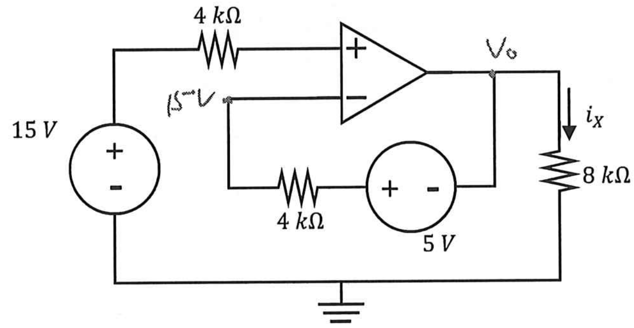
$$R_L = R_N \Rightarrow I_N = 10mA$$

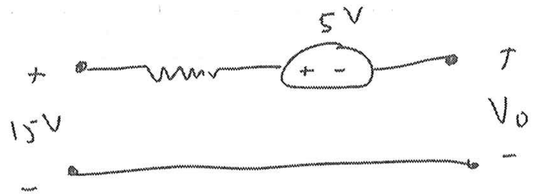
$$P_{max} = \frac{V_{Th}^2}{4R_L} = \frac{I^2 R_N^2}{4R_L} = \frac{I^2 R_N}{4} = 125\times10^{-3}$$

$$\Rightarrow R_N = \frac{125\times4\times10^{-3}}{10\times10^{-3}\times10\times10^{-3}} = 5k\Omega.$$

SECTION / AFDELING 2 [46]

Question 29 [2]			
29	Find i_X.	**29**	Bepaal i_X.

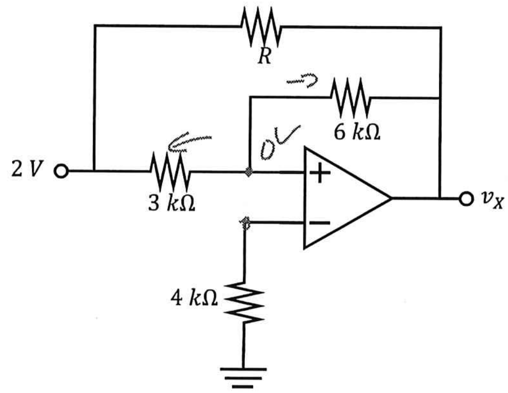




$$-15 + 5 + V_o = 0 \Rightarrow V_o = 10\,V$$

$$i_x = \frac{V_o}{8k} = 1{,}25\,mA$$

Question 30 [2]			
30	If $R \to \infty$ Ω, find v_X.	**30**	Indien $R \to \infty$ Ω, bepaal v_X.

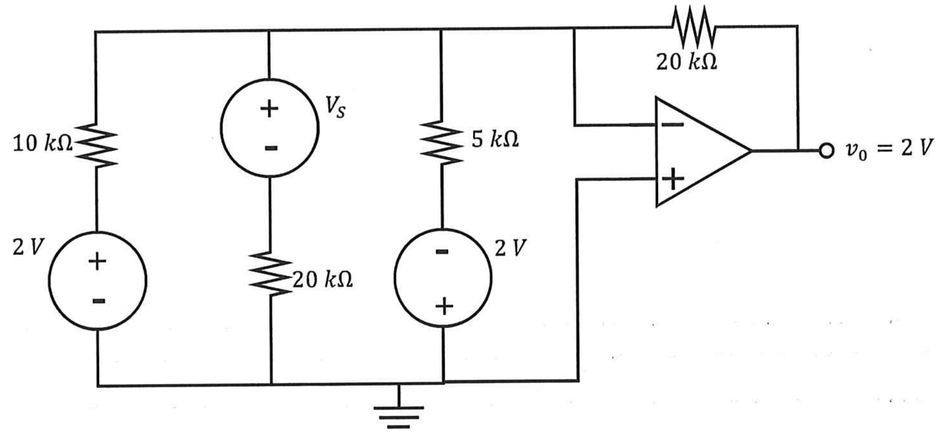


$$\frac{0-2}{3k} + \frac{0-V_x}{6k} = 0$$

$$\times 6k: \; -4 - V_x = 0$$

$$\underline{V_x = -4V}$$

Question 31 [2]			
31	Find V_S.	31	Bepaal V_S.

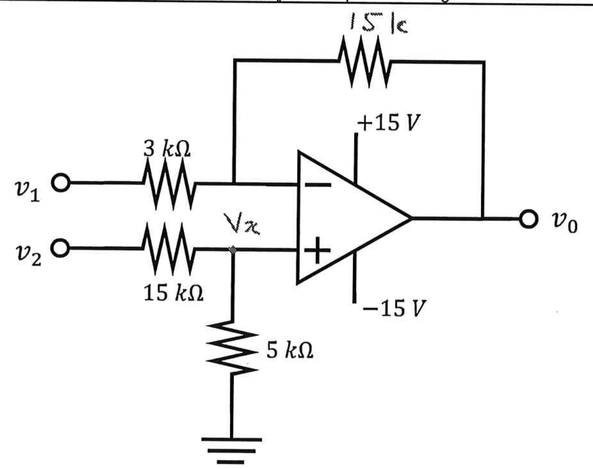


Summing : $V_o = -20k\left(\frac{2}{10k} + \frac{V_1}{20k} - \frac{2}{5k}\right)$

$-\frac{2}{20k} + \frac{2}{10k} = \frac{V_s}{20k}$

$\underline{V_s = 2V}$

Question 32 & 33 [4]			
32	If $v_1 = 0$ V and $v_2 = 5$ V, find v_o.	**32**	Indien $v_1 = 0$ V en $v_2 = 5$ V, bepaal v_o.
33	If $v_2 = 5$ V, for what value of v_1 will the output v_0 reach maximum saturation?	**33**	Indien $v_2 = 5$ V, vir watter waarde van v_1 sal die uitset v_0 maksimum versadiging bereik?



32) $V_x = \frac{5k}{5k + 15k}(5) = 1,25V$

$V_o = \left(1 + \frac{15k}{3k}\right)(1,25) = 7,5\ V$ (non-inverting)

33) $\frac{1,25 - V_1}{3k} + \frac{1,25 - V_o}{15k} = 0$

$\times 15k$ $\quad 6,25 - 5V_1 + 1,25 - V_o = 0$

$V_o = \cancel{6,25}\ 7.5 - 5V_1$

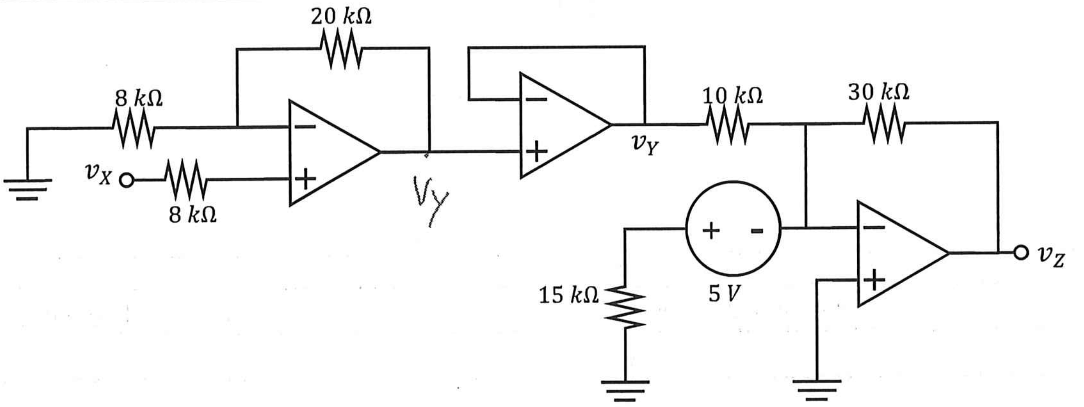
$-15 \le \cancel{6,25}\ 7.5 - 5V_1 \le 15$

$-22.5\ \cancel{-21,25} \le -5V_1 \le \cancel{8,75}\ 7.5$

$-1,75 \le V_1 \le \cancel{4,25}\ 4.5$

$\therefore V_1 = \cancel{-1,75}\ -1.5\ V.$

Question 34 & 35 [4]			
Calculate v_X and v_Z if v_Y = 7V.		Bereken v_X en v_Z indien v_Y = 7V.	
34	v_Z	**34**	v_Z
35	v_X	**35**	v_X



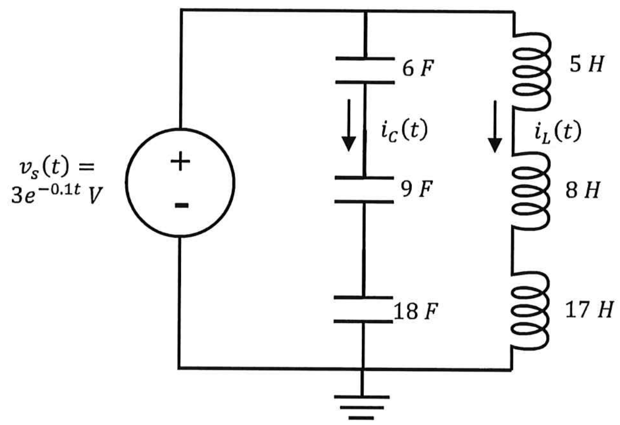
35) $V_Y = \left(1 + \frac{20k}{8k}\right) V_x$ (non-inverting)

$\therefore V_x = 2V$

34) $V_z = -30k\left(\frac{7}{10k} - \frac{5}{15k}\right)$ (summing)

$V_z = -11V$

Question 36 & 37 [4]			
At t = 10 s, what is the total current through the combination of inductors $i_L(t)$, and the total current through the combination of capacitors $i_C(t)$? Assume $i_L(0)$ = 0 A, and $i_C(0)$ = 0 A.		By t = 10 s, wat is die totale stroom deur die kombinasie van induktors $i_L(t)$, en die totale stroom deur die kombinasie van kapasitors $i_C(t)$? Aanvaar $i_L(0)$ = 0 A, en $i_C(0)$ = 0 A.	
36	i_C (10s)	**36**	i_C (10s)
37	i_L (10s)	**37**	i_L (10s)



36) $C_{eq} = 18 // 9 // 6 = 3\ F$

$$i_c(t) = C_{eq} \frac{dV_s}{dt}$$

$$= 3 \frac{d}{dt}\left(3e^{-0,1t}\right)$$

$$= -0,9 e^{-0,1t}\ A$$

$$i_c(10) = -331,09\ mA$$

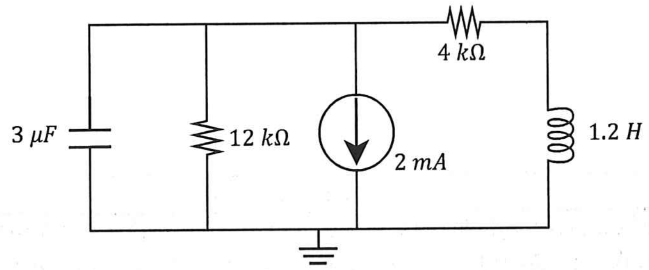
37) $L_{eq} = 5 + 18 + 17 = 30 H$

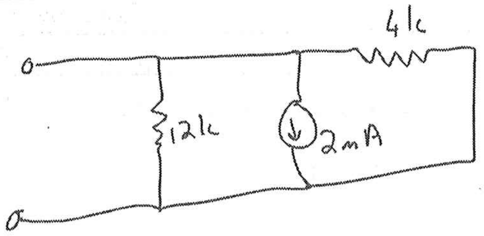
$$i_L(10) = \frac{1}{30}\int_0^{10} 3e^{-0,1t}\, dt + 0\ A$$

$$= -1\left[e^{-0,1t}\right]_0^{10}$$

$$= 632,12\ mA$$

Question 38 & 39 [4]			
Assume steady-state conditions (Direct Current) for the circuit below. What is the energy stored by the capacitor w_C and inductor w_L?		Aanvaar gestadigde toestand (Gelykstroom) vir onderstaande kringbaan. Wat is die energie gestoor deur die kapasitor w_C en induktor w_L?	
38	w_C	**38**	w_C
39	w_L	**39**	w_L





39) $I_L = -\frac{12k}{12k+4k}(2m) = -1{,}5\,mA$

$w_L = \frac{1}{2}(1{,}2)(-1{,}5\,m)^2$

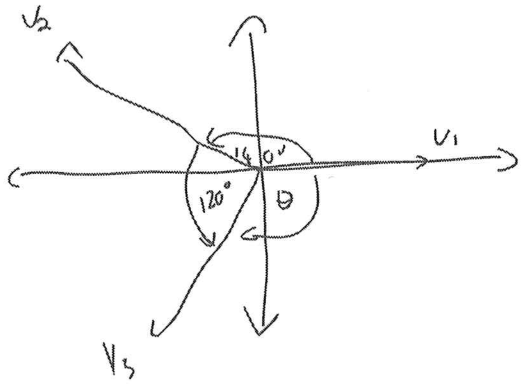
$= 1{,}35\,uJ$

38) $V_C = (-0{,}5m)(12k) = -6V$

$w_C = \frac{1}{2}(3u)(-6)^2 = 54\,\mu J$

Questions 40 & 41 [4]			
40	Sinusoids v_1, v_2 and v_3 all have the following general form: $A\cos(\omega t + \phi)$. It is known that: - v_2 leads v_1 by $140°$. - v_2 lags v_3 by $120°$. What is the phase angle θ between v_1 and v_3? $(\theta \leq \lvert 180° \rvert)$	40	Sinusgolwe v_1, v_2 en v_3 het almal die volgende algemene vorm: $A\cos(\omega t + \phi)$. Dit is bekend dat: - v_2 voor v_1 loop met $140°$. - v_2 na v_3 loop met $120°$. Wat is die fasehoek θ tussen v_1 en v_3? $(\theta \leq \lvert 180° \rvert)$
41	The following sinusoids are given: $v_1 = 2\sin(\omega t + 30°)\ V$ $v_2 = -2\cos(\omega t - 60°)\ V$ Determine the phasor value of $\bar{V}_T = \bar{V}_1 + \bar{V}_2$. Give the answer in **polar form.**	41	Die volgende sinusgolwe word gegee: $v_1 = 2\sin(\omega t + 30°)\ V$ $v_2 = -2\cos(\omega t - 60°)\ V$ Bepaal die fasor hoeveelheid: $\bar{V}_T = \bar{V}_1 + \bar{V}_2$. Gee die antwoord in **polêre vorm.**

40)



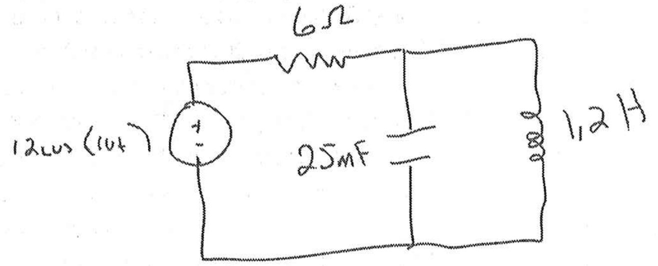
$\theta = 360° - 140° - 120°$
$= 100°$

41) $V_1 = 2\cos(\omega t + 30 - 90°) = 2\cos(\omega t - 60°)\,V$

$V_2 = 2\cos(\omega t - 60 + 180) = 2\cos(\omega t + 120°)\,V$

$\therefore V_T = 2\angle -60° + 2\angle 120°$
$= 0\,V$

Question 42 [2]	
The positive terminal of a $12\cos(10t)\ V$ voltage source is connected in series with a 6 Ω resistor. The other end of the resistor is connected in series with the parallel combination of a 25 mF capacitor and a 1.2 H inductor. The other end of the parallel capacitor and inductor combination is connected to the negative terminal of the voltage source. What is the total current through the circuit? (Assume the current exits the positive terminal of the voltage source. Give the answer in **rectangular form**.)	Die positiewe terminaal van 'n $12\cos(10t)\ V$ spanningsbron word in serie met 'n 6 Ω resistor gekoppel. Die ander punt van die resistors is in serie met die parallel kombinasie van 'n 25 mF kapasitor en 1.2 H induktor gekoppel. Die ander punt van die parallele kapasitor en induktor kombinasie is aan die negatiewe terminaal van die spanningsbron gekoppel. Wat is die totale stroom in die kringbaan? (Aanvaar stroom vloei uit die positiewe terminaal van die spanningsbron. Skryf die antwoord in **reghoekige vorm**.)
42 $\bar{I}_T$	**42** $\bar{I}_T$



$$Z_{eq} = 6 + \frac{1}{j(10)(25m)} \,//\, j(10)(1,2)$$

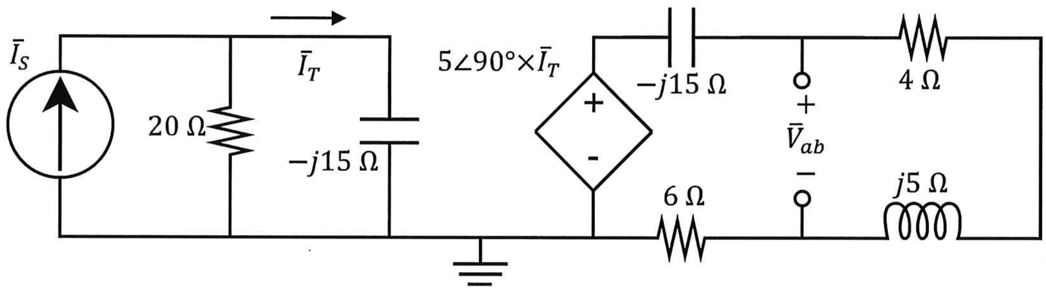
$$= 6 + (-j4)\,//\,j12$$

$$= 6 - j6$$

$$\underline{I}_T = \frac{12\angle 0^\circ}{6 - j6}$$

$$\underline{I}_T = 1 + j\ A$$

Question 43 & 44 [4]			
It is given that $\bar{I}_T = 2\angle 0°A$. Determine $\bar{I}_S$ and $\bar{V}_{ab}$.		Dit word gegee dat $\bar{I}_T = 2\angle 0°A$. Bepaal $\bar{I}_S$ en $\bar{V}_{ab}$.	
43	$\bar{I}_S$ **(rectangular form.)**	**43**	$\bar{I}_S$ **(reghoekige vorm)**
44	$\bar{V}_{ab}$ **(rectangular form.)**	**44**	$\bar{V}_{ab}$ **(reghoekige vorm)**



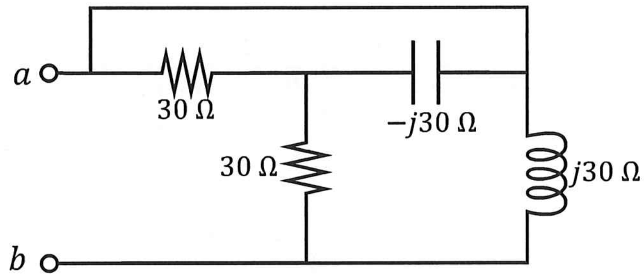
43) $2\angle 0° = \dfrac{20}{20 - j15} I_S$

$\therefore \underline{I_S = 2 - j1{,}5 \text{ A}}$

44) $V_{ab} = \dfrac{4 + j5}{4 + j5 + 6 - j15} (5\angle 90° \times 2\angle 0°)$

$\underline{= -4{,}5 - j0{,}5 \text{ V}}$

Question 45 [2]			
45	Calculate Z_{ab}. **(Rectangular)**	45	Bereken Z_{ab}. **(Reghoekig)**

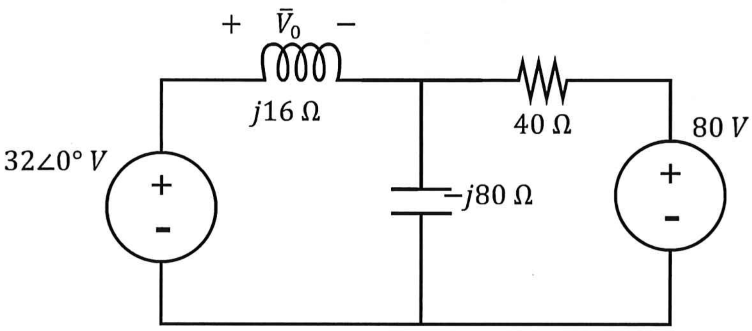


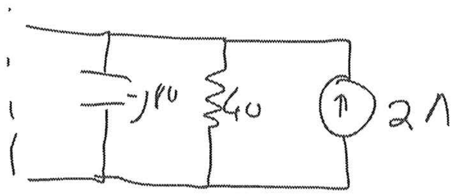
$$Z_{ab} = (30 // -j30 + 30) // j30$$

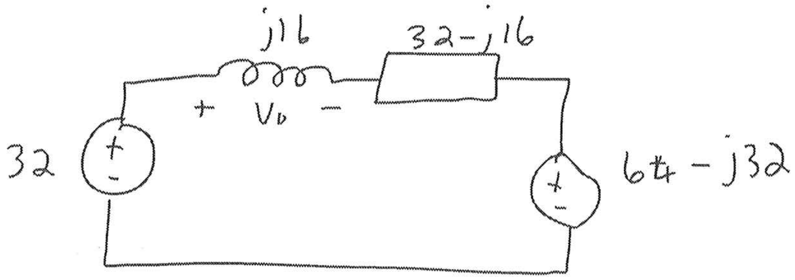
$$= (45 - j15) // j30$$

$$= \underline{18 + j24\ \Omega}$$

Question 46 [2]			
46	Determine $\bar{V}_0$ using source transformation (Write the answer in **rectangular form**.)	**46**	Bepaal $\bar{V}_0$ m.b.v. brontransformasie. (Skryf die antwoord in **reghoekige vorm**.)







$$-32 + 32I + 64 - j32 = 0$$

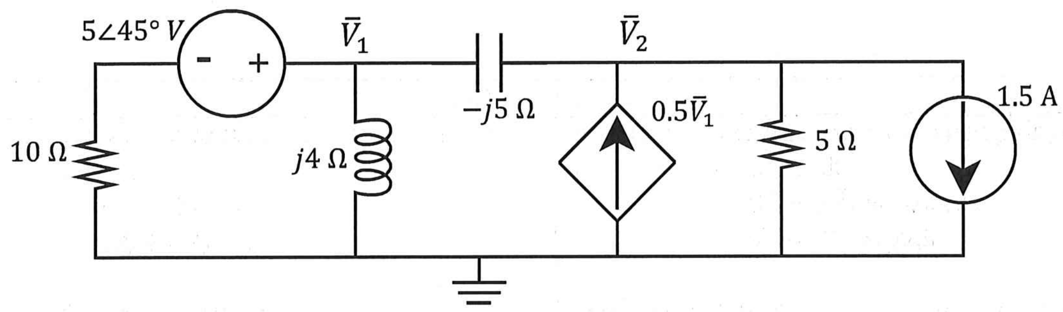
$$\therefore I = -1 + j$$

$$V_0 = (-1+j)(j16)$$

$$= -16 - j16$$

Question 46 to 49 [4]	
Use **nodal** analysis to set up two equations of the following form: $$a\bar{V}_1 + b\bar{V}_2 = 10\angle 135^\circ$$ $$c\bar{V}_1 + d\bar{V}_2 = j7.5$$ Give the coefficients in **rectangular form**.	Gebruik **node**-analise om twee vergelykings van die volgende vorm op te stel: $$a\bar{V}_1 + b\bar{V}_2 = 10\angle 135^\circ$$ $$c\bar{V}_1 + d\bar{V}_2 = j7.5$$ Bepaal die koëffisiënte in die **reghoekige vorm**.

46	a	**47**	b	**46**	a	**47**	b
48	c	**49**	d	**48**	c	**49**	d



$$\frac{V_1 - 5\angle 45^\circ}{10} + \frac{V_1}{j4} + \frac{V_1 - V_2}{-j5} = 0$$

$\times j20$: $j2V_1 - 10\angle 135^\circ + 5V_1 - 4V_1 + 4V_2 = 0$

$$\underline{(1 + j2)V_1 + 4V_2 = 10\angle 135^\circ}$$

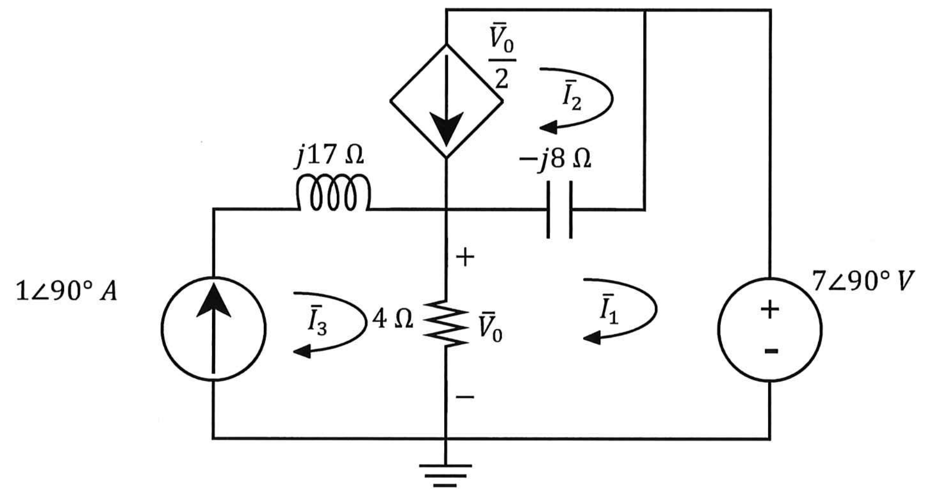
$$\frac{V_2 - V_1}{-j5} + \frac{V_2}{5} - 0.5V_1 + 1.5 = 0$$

$\times -j5$: $V_2 - V_1 - jV_2 + j2.5V_1 - j7.5 = 0$

$$\underline{(-1 + j2.5)V_1 + (1 - j)V_2 = +j7.5}$$

Question 50 to 53 [4]	
Use mesh analysis to set up two equations of the following form: $e\bar{I}_1 + f\bar{I}_2 = -j3$ (Loop 1) $g\bar{I}_1 + h\bar{I}_2 = j2$ Give the coefficients in **rectangular form**.	Gebruik lusanalise om twee vergelykings van die volgende vorm op te stel: $e\bar{I}_1 + f\bar{I}_2 = -j3$ (Lus 1) $g\bar{I}_1 + h\bar{I}_2 = j2$ Bepaal die koëffisiënte in die **reghoekige vorm**.

50	*e*		**51**	*f*		**50**	*e*		**51**	*f*	
52	*g*		**53**	*h*		**52**	*g*		**53**	*h*	



$L1:\quad 7\angle 90° + (4-j8)I_1 - 4I_3 + j8I_2 = 0$

$I_3 = 1\angle 90°$

$(4-j8)I_1 - 4\angle 90° + j8I_2 = -7\angle 90°$

$\underline{(4-j8)I_1 + j8I_2 = -j3}$

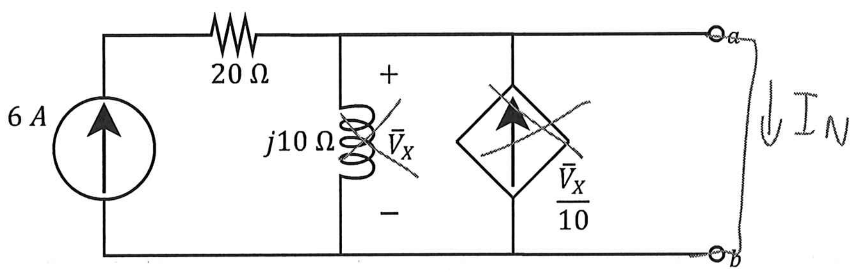
$V_0 = 4I_3 - 4I_1$ but $I_2 = -\frac{V_0}{2}$

$V_0 = -2I_2$

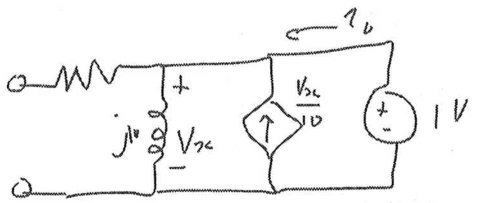
$-2I_2 = 4\angle 90° - 4I_1$

$\underline{2I_1 - I_2 = 2j}$

Question 54 & 55 [4]			
Calculate the Norton equivalent w.r.t. terminals *a*-*b*.		Bereken die Norton ekwivalent m.v.t. terminale *a*-*b*.	
54	$\bar{I}_N$ **(rectangular form.)**	**54**	$\bar{I}_N$ **(rectangular form.)**
55	Z_N **(rectangular form.)**	**55**	Z_N **(rectangular form.)**



$I_N = 6A$



$i_o = \frac{1}{j10} - \frac{1}{10}$

$= -0,1 - j0,1$ A

$\therefore Z_N = \frac{1}{i_o} = -5 + j5\ \Omega$

PRACTICE PAGE / OEFENBLAD 1

Assessment ID	2	0	1	8	E	B	N	1	1	1	E	1		

Make use of 5 significant digits for all answers involving irrational numbers.
Maak gebruik van 5 beduidende syfers vir alle antwoorde bestaande uit irrasionele getalle.

SECTION 1

1	$t =$ 60	s
2	$cost =$ 3.50	R
3	$P_{Vd} =$ 0.2	W
4	$P_{R1} =$ 0.4	W
5	$V_{R1} =$ 10	V
6	$R =$ 40	Ω
7	$R_{cd} =$ 35	kΩ
8	$a =$ 10	*No Unit Required*
9	$b =$ −8	
10	$c =$ −2	
11	$d =$ −1.5	
12	$e =$ 18	
13	$f =$ −25	
14	$g =$ −1	
15	$h =$ 1	
16	$V_0 =$ 10	V
17	$\Delta_M =$ −36	*No Unit Required*
18	$V_I =$ 3.225	V
19	$V_{CE} =$ 1.525	V
20	$v_x =$ −3.2	V

PRACTICE PAGE / OEFENBLAD 2

Assessment ID	2	0	1	8	E	B	N	1	1	1	E	1		

Make use of 5 significant digits for all answers involving irrational numbers.
Maak gebruik van 5 beduidende syfers vir alle antwoorde bestaande uit irrasionele getalle.

21	$v_{y(1A)}$ = −3.75	V
22	$v_{y(10V)}$ = −2.5	V
23	V_{Th} = 1.2	V
24	R_{Th} = 300	Ω
25	I_N = 2.75	mA
26	R_N= 16	kΩ
27	I_N = 10	mA
28	R_N= 5	kΩ

SECTION 2

29	i_X = 1.25	mA
30	v_X = −4	V
31	V_S= 2	V
32	v_0= 7.5	V
33	v_1= −1.5	V
34	v_Z= −11	V
35	v_X = 2	V
36	i_C (10s) = 632.12	mA
37	i_L (10s) = −331.09	mA
38	w_C = ~~E~~ 54	μJ
39	w_L = 1.35	μJ

PRACTICE PAGE / OEFENBLAD 3

Assessment ID	2	0	1	8	E	B	N	1	1	1	E	1		

Remember:
Write a complex number in:

- rectangular form as $a \pm j\,b$ where j is at the **start** of the imaginary number.
- polar form as $a\angle\theta°$ where the angle sign $\angle$ is clear, and you add the degree unit after the angle).

(See examples on page 2.)

You may write a complex number either in polar or in rectangular format. Both answers are marked correct.

40	$\theta =$ 100°		
41	$\bar{V}_T =$ 0∠0°	**(polar)**	V
42	$\bar{I}_T =$ 1 + j	**(rectangular)**	A
43	$\bar{I}_S =$ 2 − j1.5	**(rectangular)**	A
44	$\bar{V}_{ab} =$ −4.5 − j0.5	**(rectangular)**	V
45	$Z_{ab} =$ 18 + j24	**(rectangular)**	Ω
46	$\bar{V}_0 =$ −16 − j16	**(rectangular)**	V
47	$a =$ 1 + j2	**(rectangular)**	*No unit required.*
48	$b =$ 4	**(rectangular)**	
49	$c =$ −1 + j2.5	**(rectangular)**	
50	$d =$ 1 − j	**(rectangular)**	
51	$e =$ 4 − j8	**(rectangular)**	
52	$f =$ j8	**(rectangular)**	
53	$g =$ 2	**(rectangular)**	
54	$h =$ −1	**(rectangular)**	
55	$\bar{I}_N =$ 6	**(rectangular)**	A
56	$Z_N =$ −5 + j5	**(rectangular)**	Ω